

Coffee shop management system Adriel Nguyen

Problem Statement:

Coffee shops are ready to open in the morning, but 30 customers need to wait in line for purchases.

Managing a coffee shop is not easy, from closely keeping track of supplies bought, customer progress, and employees' sales. The biggest problem that impacts the company is when a manager does not know when to order the types of supplies needed for employees to make drinks for the customers.

Improper coffee shop management directly leads to consistent customer complaints. Aside from this, it prevents opportunities for the customer to purchase the service in terms of ending up wasting their time, they do not know that such customer benefits exist, or perhaps their manager cannot see how much employees' sales increase during their working hours.

Objectives of the system:

The purpose of the system helps the coffee shop serve coffee strategically to the customer under time constraints, which would accelerate faster flow, more productivity, and easier control. For instance, it would let customers preorder a menu when waiting, split into lanes, and waiting numbers to call; all these three activities are happening at the same time. Users can keep track of their membership progress, which would let them know when their drinks are ready to pick up.

The systems will help coffee managers to visualize current inventory and to order ingredients accordingly.

Typical customers

Office workers, freelancers, researchers, students, real estate agents, ...

System Requirements:

- Managers can plan to order the necessary ingredients/materials.
- Employees can keep track of user information, their orders, and exchange membership points to buy drinks, set up a pre-order process, activate self-service orders, and manage waiting numbers.
- Customers can view access membership, order drinks, and maybe use a bar code to pay for the drink.

Project planning and development

1. Software:

Front-end: Node.js, CSS

Back end: Python, Java

Database: MySQL / MariaDB

Framework use: SymPy, NumPy, Apache NetBeans

1. Hardware: Personal laptop

2. No additional plugins needed for networks

Week 1: Writing pseudocode, gathering information, and reading books relating to data structure, database, and preparing presentations properly. GitHub commit and to-do lists.

Week 2: Make sure backend and frontend can communicate with each other; the backend can update and change in real time

Week 3: Customer progress register, track supplies mode, employee sales record,

Week 4: Adds on other functionality, preordering, self-order lanes, and ticket number

Week 5: GUI and interaction, add-ons and code revision, update and record the demo.

Week 6: Run multiple live tests, fix mistakes

Week 7: Test edge cases and note abnormal changes

Week 8: Record final demo and presentation